

Systemic sclerosis

Spatial multiomics decipher fibroblast–macrophage dynamics in systemic sclerosis

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ABSTRACT

Objectives: Stromal-immune crosstalk shapes the pathogenic microenvironment of systemic sclerosis (SSc), but the spatial regulatory networks underlying fibrogenesis remain poorly defined. We aimed to explore tissue organisation and cell coordination in SSc skin, providing spatiotemporal insights into disease mechanisms and bridging the gap between omics discovery and precision medicine.

Methods: We performed spatial transcriptomics on skin biopsies from 10 patients with diffuse cutaneous SSc and 4 healthy controls using the 10× Visium platform. These findings were confirmed using higher-resolution Stereo-seq transcriptomics, spatial proteomics, and single-cell RNA sequencing data from patients with SSc, SSc mouse models, and wound-healing reindeer models. *In vivo* and *in vitro* studies were conducted to validate the key regulatory pathways.

Results: Fourteen skin biopsies were analysed, revealing significant expansion of fibrotic niches enriched with fibroblasts and macrophages in SSc, correlating with clinical severity. We revealed disease-specific cell states of fibroblasts and macrophages and evaluated their spatial

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dependency and cell–cell communication. Stratification based on signature genes enabled the identification of patients with SSc with progressive disease and treatment-nonresponsive phenotype. ACKR3 (a CXCL12 decoy receptor) was selectively expressed in myofibroblast progenitors, which diminished during differentiation towards mature myofibroblast, potentially serving to regulate CXCL12/CXCR4-mediated proinflammatory macrophage recruitment. Inhibition of CXCR4 attenuated skin and lung fibrosis in experimental fibrosis mouse models.

Conclusions: Our spatially resolved atlas uncovered dynamic fibroblast–macrophage interplay as a hallmark of fibrotic niche expansion. These findings offer spatiotemporal insights into disease mechanisms and pave the way for advanced mechanistic and therapeutic studies, bridging the gap between omics discovery and precision medicine.

WHAT IS ALREADY KNOWN ON THIS TOPIC

- Translating omics discoveries into clinically actionable strategies, such as biomarker-driven patient stratification or mechanism-based target identification, remains a critical unmet challenge in systemic sclerosis (SSc).
- While single-cell omics have revolutionised our understanding of cellular heterogeneity in SSc, integrating spatial context is essential to unravel tissue architecture and cell–cell coordination driving fibrogenesis.
- Although stromal-immune crosstalk, particularly fibroblast–macrophage interactions, has been implicated in tissue fibrosis, the niche-specific spatiotemporal dynamics of these interactions in SSc are not fully resolved.

WHAT THIS STUDY ADDS

- Integrated spatial omics elucidated the fibrotic microenvironment in SSc skin, showing spatial organisation, transcriptional profiles, dynamic interactions, and their links with clinical outcomes.
- Cross-tissue and cross-species analyses identified a core differentiation axis from myofibroblast progenitors to terminally differentiated myofibroblasts, highlighting shared mechanisms in fibrotic remodelling.
- In-depth cell communication analysis uncovered fine-tuned regulatory events involving fibroblasts and macrophages, which may potentially drive the progression of SSc.
- A myofibroblast progenitor-specific chemokine regulatory mechanism is diminished with the disease progression, implicating fibroblast–macrophage crosstalk in sustaining fibrosis and nominating targetable pathways for intervention.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- Our study deciphers the microenvironment driving SSc progression, identifies potential therapeutic targets, and highlights the potential of omics-based stratification to predict treatment response, bridging the gap between omics discovery and precision medicine in clinical practice.

INTRODUCTION

Systemic sclerosis (SSc), an orphan autoimmune disease, is characterised by the progressive fibrosis affecting skin and multiple organs, holding the highest mortality among all rheumatic diseases [1]. The development of effective therapeutic strategies for SSc is impeded by a lack of comprehensive understanding of its pathogenesis and the limited translatability of current pre-clinical models. Importantly, studying SSc as a prototypical systemic fibrotic disease could yield new insights into the pathogenesis of fibrotic diseases and aid in the discovery of novel therapeutic approaches.

SSc is characterised by a complex interplay of features that include vasculopathy, inflammation, and fibrosis. Within this triad, fibroblasts have been recognised as pivotal effector cells in the production of extracellular matrix, largely contributing to developing fibrotic tissue remodelling in SSc. Fibroblasts sense and receive signalling from the neighbouring cells, which are conveyed and converged to the shared and distinct profibrotic pathways [2]. The coordination of fibroblasts with macrophages can shape their cellular response, therefore orchestrating the initiation or resolution of fibrosis across tissues [3–9]. Beyond the conventional macrophage polarisation paradigm, new populations of macrophages have been linked with tissue repair and fibrotic remodelling [3–7,10]. For instance, phenotypically distinct macrophages (LYVE1^{lo}MHCII^{hi} and LYVE1^{hi}MHCII^{lo}) manifested specific niche-dependent functional roles in pulmonary and cardiac fibrosis [10]. However, it remains largely unexplored whether and how the spatial neighbourhood in the tissue architecture affects the crosstalk between fibroblast and macrophage subsets, potentially driving fibrotic remodelling in SSc.

Recent studies using single-cell RNA sequencing (scRNA-seq) have provided an unprecedented opportunity to study the molecular and cellular mechanisms of SSc at single-cell resolution [11–15]. Pioneering studies have characterised fibroblast types and states in SSc [11–13,15]. While these studies have provided valuable insights, they are significantly constrained by the lack of spatial resolution due to the reliance on disaggregated tissue analysis. We recently demonstrated cellular niches centred around endothelial cells (ECs) or fibroblasts in SSc using a panel of up to 40 markers through multiplex spatial proteomics [16,17]. Thus, a spatially resolved characterisation of genome-wide transcriptomics of SSc is needed.

In this work, we profiled human skin from patients with SSc and healthy controls using spatial transcriptomics based on 10× Visium formalin-fixed paraffin-embedded platform and spatial-enhanced resolution omics sequencing (Stereo-seq) [18]. To increase resolution, we integrated 2 publicly available scRNA-seq data sets, creating a human skin transcriptome atlas and mapping cell type information onto spatial regions. Validation included higher-resolution spatial profiling (eg, imaging mass cytometry [IMC]) and time-course experimental skin fibrosis. These methods characterised cellular heterogeneity and pathologic interactions between fibroblast and macrophage subsets, identifying potential therapeutic targets through disruption of disease-associated cellular niches. This work connects multi-omics findings to clinically relevant strategies for SSc.

METHODS

A detailed description of all materials and methods are provided in [Supplementary Methods](#). All human studies were approved by the ethical committee of the Medical Faculty of

Fudan University. All patients and controls signed an informed consent form approved by the local institutional review board. Statistical measures, including the number of samples, descriptive statistics (median and IQR), and significance, are reported in the figure and figure legends. *P* values of <.05 were considered statistically significant.

RESULTS

Profiling of SSc and healthy skin with spatially resolved transcriptomics

We collected 14 human skin biopsies from 10 patients with diffuse cutaneous SSc with a median disease duration of 8 months since the first non-Raynaud phenomenon (RP), as well as from 4 sex-matched and age-matched normal controls (Fig 1A; Supplementary Table S1). For each biopsy, we generated spatially resolved transcriptomics using the 10× Visium platform and retained 13,204 high quality spots, with a median detection of 2265 genes per spot after removing low quality data (Supplementary Fig S1A-D; Supplementary Table S2). To enhance spatial resolution, we also performed Stereo-seq on 4 of these skin samples, including 3 from patients with SSc and 1 from a healthy donor. After controlling data quality, we retained 40,566 spots across these samples. We additionally analysed 2 scRNA-seq data sets generated from skin tissues of patients with SSc (*n* = 104) and healthy donors (*n* = 64) [11,12] (Fig 1B,C; Supplementary Fig S2). We deconvoluted the cell type compositions of each spot in 10× Visium data using the scRNA-seq as reference. The estimated cell type abundances from spatial transcriptomics were generally colocalised with their respective marker genes and locations (Fig 1D). For instance, keratinocytes were found localised within the epidermis and surrounding the hair follicle. We also inferred the pathway activity for each spot and observed a strong association between marker expression and some well-known targetable pathways, such as Janus kinase signal transducer and activator of transcription protein (JAK-STAT) [19–22], transforming growth factor (TGF)- β [2], and interleukin (IL)-6 [23–25] using PROGENy pipeline (Fig 1D). We next transferred the cell type annotation from scRNA-seq to Stereo-seq through data integration and observed a similar cellular distribution as 10× Visium data (Fig 1E,F). Collectively, our analyses of spatial transcriptomics and scRNA-seq data enables us to map cell type information to spatial for skin samples.

Characterisation of tissue organisation in SSc and healthy skin

To characterise the cellular colocalisation events in human skin, we clustered the spots based on estimated cell type proportions and identified 12 cell type niches (Fig 2A; Supplementary Fig S3A). Visualisation of the cell type abundances and marker genes of these niches revealed that they were strongly associated with the corresponding cell types (Fig 2B,C; Supplementary Fig S3B,C). Furthermore, we observed a significant increase in the proportions of 3 niches (0, 1, and 4) in the SSc skin compared with those in healthy skin (Fig 2D,E; Supplementary Fig S3D,E). Notably, niche 0 was infiltrated with a high prevalence of fibroblasts, macrophages, conventional dendritic cells (cDCs), and mast cells, indicating a potential immune microenvironment within a fibrogenic niche. Niche 1 was characterised by an enrichment of ECs, pericytes, macrophages, cDCs, and mast cells, suggesting a perivascular niche associated with immune cell infiltration in SSc skin. Niche 4 was distinguished by a significant presence of plasma cells, melanocytes, and neural cells.

Decoupler pathway activity analyses for each niche demonstrated the highest activity of the TGF- β signalling pathway in fibroblast-dominant niche 0 (Fig 2F). Niche 9, composed of various immune cells, showed an increased trend in proportion and exhibited the strongest activity of JAK-STAT signalling. KEGG pathway enrichment analysis based on marker genes also revealed high activity in the TGF- β signalling in niche 0 (Supplementary Fig S3F). These findings supported the essential roles of TGF- β and JAK-STAT signalling in the pathogenesis of SSc as expected [2], suggesting a convergence of multiple signalling activities within the structural organisation in SSc fibrosis.

Notably, we observed colocalisation of fibroblasts and macrophages in our 10× Visium and Stereo-seq data (Fig 2G,H). To validate these results, we performed IMC and identified the cellular neighbourhood (CN) community as described previously [17]. We observed similar niches comprising stromal cells and macrophages in CN8, while we found CN5 containing vascular ECs and immune cells, such as macrophages (Supplementary Fig S4A-C).

We then correlated the niche proportions detected in 10× Visium data with clinical data across the SSc samples and observed a positive Spearman correlation between the proportion of niche 0 and the current total modified Rodnan skin score (mRSS) (r = 0.733; P = .023), C-reactive protein levels (r = 0.750; P = .066), and platelet counts (r = 0.697; P = .031), which are interpreted as clinical indicators for disease severity or activity of SSc (Fig 2I). Additionally, we observed a significant negative correlation between the proportion of niche 1 and the non-RP disease duration (r = -0.756; P = .014). No correlations were observed between niche proportion and clinical outcomes, such as forced vital capacity (FVC)% predicted or diffusing capacity of the lungs for carbon monoxide% predicted (Supplementary Fig S3G). In summary, our spatial multiomics revealed a distinct structural organisation associated with disease severity and activity, suggesting a potential coordinated regulation of immune-stromal interactions that may drive SSc progression.

Characterisation of fibroblast subclusters in healthy and SSc skin

We next clustered the scRNA-seq data for fibroblasts and identified 10 distinct subclusters. Consistent with source studies [11,12], we found the corresponding fibroblast populations, including *PCOLCE2*⁺/*LGR5*⁺ (Fib1), *MYOC*⁺/*C3*⁺/*APOD*⁺ (Fib3), *CCL19*⁺/*C7*⁺/*APOE*⁺ (Fib4), *SFPR2*⁺/*WIF1*⁺/*APCDD1*⁺ (Fib6), *COCH*⁺/*SPARCL1*⁺ (Fib7), *COMP*⁺/*SFPR2*⁺/*PRSS23*⁺ (Fib9), and *IGFBP2*⁺/*ANGPTL7*⁺ (Fib10) (Fig 3A,B; Supplementary Fig S5A). Additionally, Fib5 was characterised by the expression of *FOS* and *JUNB*, which were components of activator protein 1 known as the important transcription factors (TFs) in SSc fibrosis [26,27]. Fib1 demonstrated the upregulation of *SCARA5*, which has been recently identified as a marker for myofibroblast progenitor in human myocardial and kidney fibrosis [28–30]. Fib2 and Fib9 were characterised by the expression of periostin (*POSTN*), encoding an extracellular matrix protein that can amplify TGF- β signalling [31,32]. In addition, Fib2 also exhibited a high expression of myofibroblast hallmark *ACTA2* (Fig 3B). Moreover, we observed a significant reduction of myofibroblast progenitor Fib1 and an increase of myofibroblast Fib2 and Fib9 in SSc skin, compared with those in healthy skin (Fig 3C).

Interestingly, Fib2 was characterised by high TF activity of the Hedgehog transcription factor *GLI2* (Supplementary Fig S5B). This convergent driver can mediate Hedgehog pathways

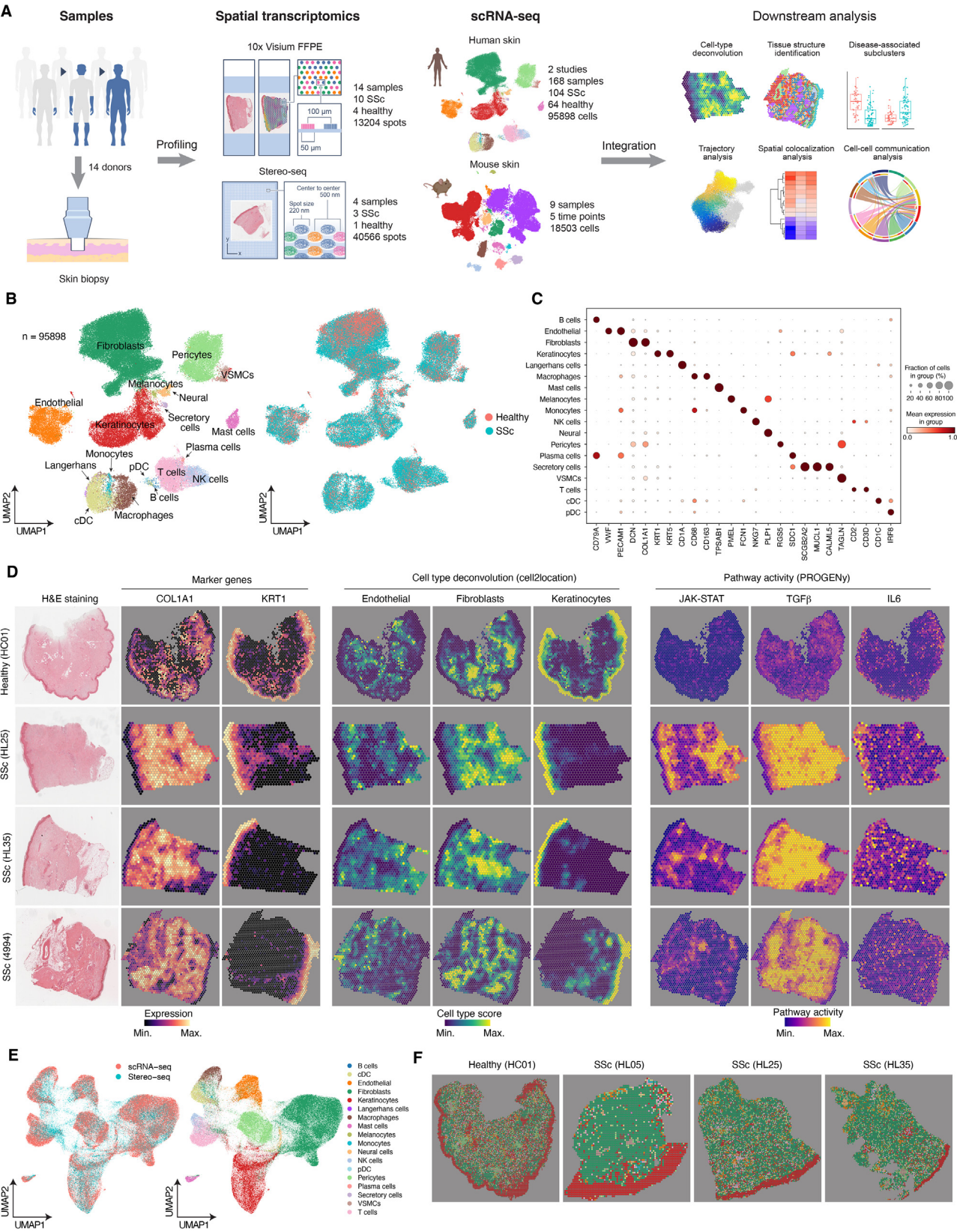


Figure 1. Spatial profiling of human skin. (A) Overview of experimental design and analysis workflow. (B) Uniform manifold approximation and projection (UMAP) showing integrated single-cell RNA sequencing (scRNA-seq) data sets as coloured by cell types (left) and conditions (right). Systemic sclerosis (SSc) skin, n = 104; healthy skin, n = 64. (C) Dotplot showing gene expression for each cell type. The colours refer to the mean expression of the genes and the dot sizes the proportion of cells expressing the marker genes. (D) Hematoxylin and eosin staining and the corresponding visualisation maps of based on x10 Visium (SSc skin, n = 10; healthy skin, n = 4). Left: *COL1A1* and *KRT1* gene expressions in spatial transcriptomics data. Middle: representative cell types estimated by cell2location. Right: visualisation of pathway activity estimated by their target genes in spatial space. (E) Integration of scRNA-seq and Stereo-seq data coloured by protocols (left) and cell types (right). (F) Visualisation of cell types in spatial space based on Stereo-seq data. SSc skin, n = 3; healthy skin, n = 1. The experiments were performed on adjacent tissue sections to those used for x10 Visium.

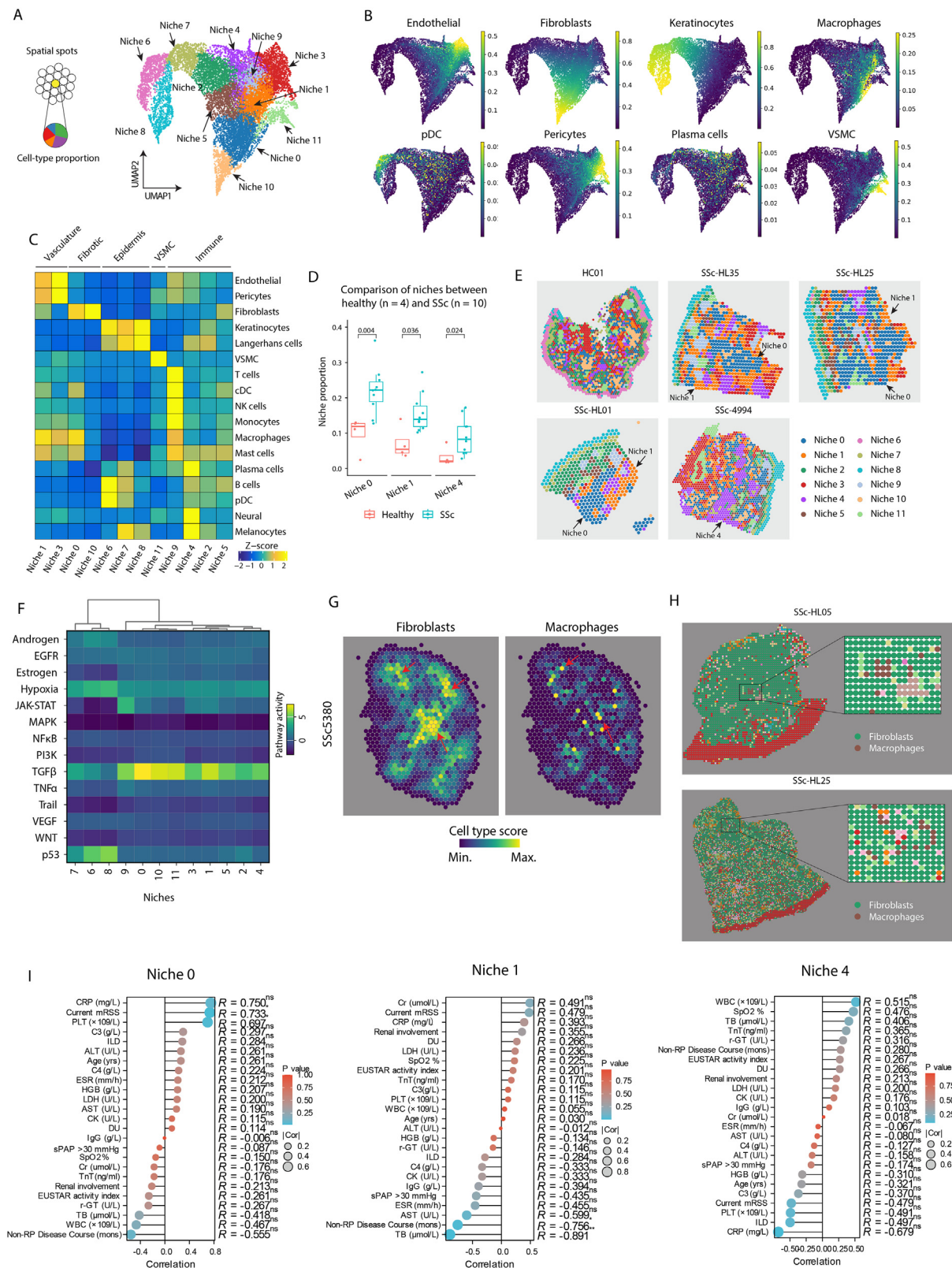


Figure 2. Characterisation of cell type niches in systemic sclerosis (SSc) and healthy skin. (A) Uniform manifold approximation and projection (UMAP) showing the integrated spatial data as coloured by cell type niches. (B) Visualisation of cell type proportion in UMAP space. (C) Heatmap showing the enrichment of cell types in niches. Each column represents a niche and each row a cell type. Colours refer to the z-score—transformed proportions of each cell type across different niches. (D) Boxplot comparing the proportion of each niche between SSc and healthy skin samples. Each dot represents a sample. P values are calculated using the Wilcoxon rank-sum test (unpaired; 2 sides). (E) Visualisation of cell type niches in spatial space for the healthy and SSc samples using ×10 Visium data. (F) Heatmap showing pathway activity for each niche. (G) Visualisation of fibroblasts and macrophages in spatial space using ×10 Visium data. (H) Visualisation of cell types in spatial space using Stereo-seq data. (I) Correlations between the proportions of the indicated niches with clinic statistics across all subjects with SSc. Statistical significance was determined by the Spearman test for a 2-variable correlation. The correlation coefficient r is indicated in figure (* $P < .05$). ALT, alanine aminotransferase; AST, aspartate aminotransferase; C4, complement component 4; CK, creatine kinase; Cr, creatinine; CRP, C-reactive protein; C3, complement component 3; DU, digital ulcer; ESR, erythrocyte sedimentation rate; EUSTAR, European Scleroderma Trials and Research Group; HGB, haemoglobin; IgG, immunoglobulin G; ILD, interstitial lung disease; LDH, lactate dehydrogenase; mRSS, modified Rodnan skin score; PLT, platelet count; r-GT, gamma-glutamyl transferase; RP, Raynaud phenomenon; sPAP, systolic pulmonary arterial pressure; SpO₂, oxygen saturation; TB, total bilirubin; TnT, troponin T; WBC, white blood cell.

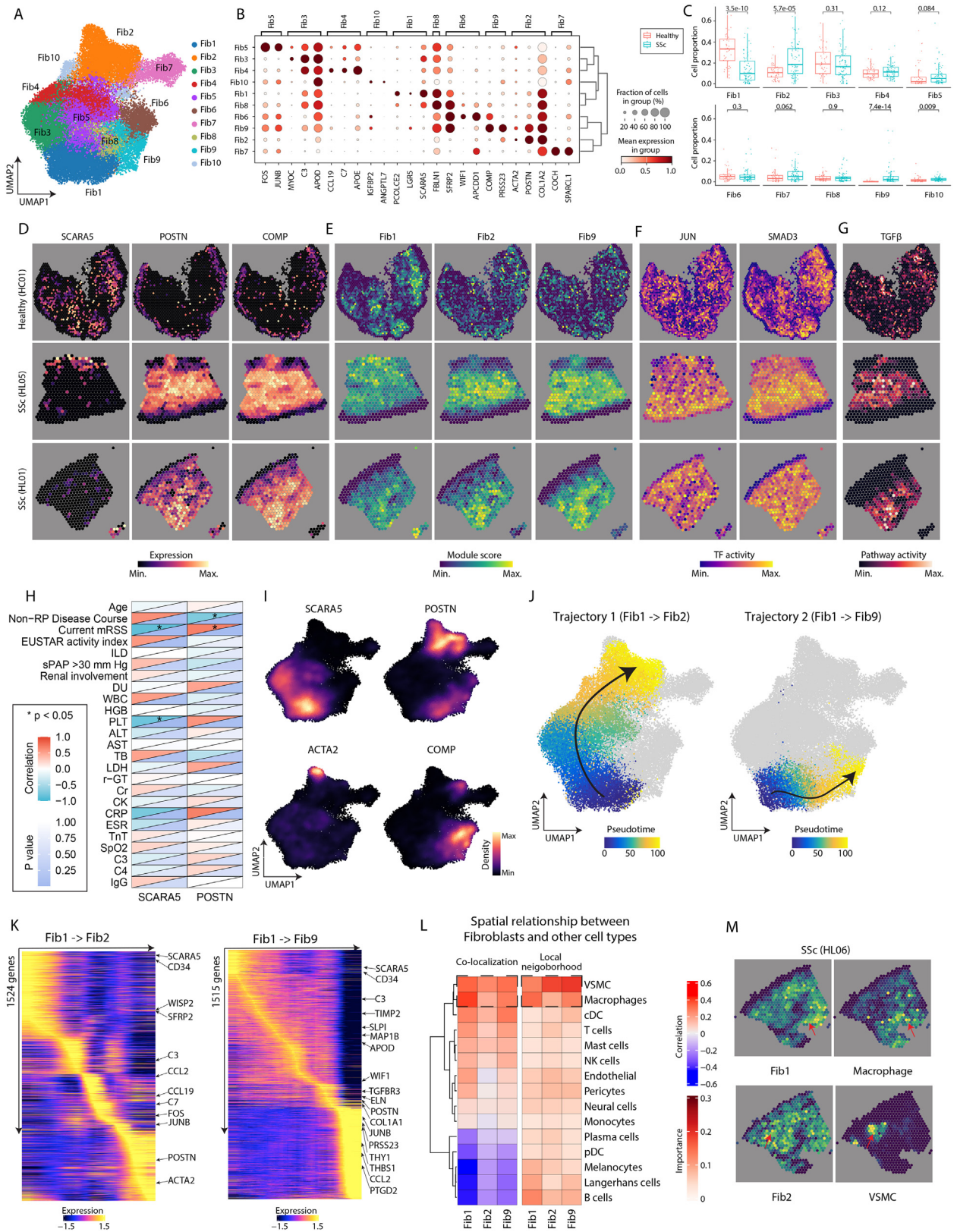


Figure 3. Characterisation of fibroblast subpopulations. (A) Uniform manifold approximation and projection (UMAP) showing subpopulations of fibroblasts based on single-cell RNA sequencing (scRNA-seq) data. (B) Dot plot showing the marker genes of each subset. (C) Bar graph comparing the proportion of each fibroblast subset between healthy and systemic sclerosis (SSc) samples. Each dot represents a sample. *P* values are calculated using the Wilcoxon rank-sum test (unpaired; 2 sides). (D) Visualisation of *SCARAS*, *POSTN*, and *COMP* expression in spatial space. (E) Visualisation of module score of Fib1, Fib2, and Fib9 in spatial space. (F) Visualisation of transcription factor (TF) activity of JUN and SMAD3 in spatial space. (G) Visualisation of transforming growth factor (TGF)- β pathway activity in spatial space. (H) Heatmap showing correlations between expressions of *SCARAS* and *POSTN* and different clinical outcomes of SSc. The colours in the upper left square represent Spearman *r* value, while those in the lower right square represent *P* values ($*P < .05$). (I) Expression of *SCARAS*, *POSTN*, *ACTA2*, and *COMP* overlaying with UMAP. (J) UMAP showing the differentiation trajectory from Fib1 to Fib2 (left) and Fib1 to Fib9 (right). Colours refer to pseudotime points. (K) Heatmap showing the gene expression dynamics during the trajectory from Fib1 to Fib2 (left) and Fib1 to Fib9 (right). (L) Colocalisation between major cell types and fibroblast subclusters (Fib1, Fib2, and Fib9) as measured within the same spot (left) and neighbouring spots (right). (M) Visualisation of the abundance of macrophages and

in activating fibroblast and promoting fibrosis in SSc [27,33]. Fib9 was marked by *SMAD3*, *SMAD4*, and *JUN*, all of which are TGF- β downstream pathways to promote fibrosis. We also analysed a bulk RNA-seq dataset from the prospective registry of early systemic sclerosis (PRESS) cohort [34] and observed the significant increase of *POSTN* and decrease of *SCARA5* in SSc skin compared with healthy skin (Supplementary Fig S5C). An inverse correlation was also observed in the gene expressions between *SCARA5* and *POSTN*. Overall, these findings further supported the potential involvement of *SCARA5*⁺ and *POSTN*⁺ fibroblasts in the pathogenesis of SSc.

We next analysed our spatial transcriptomics data and found that *SCARA5* was significantly downregulated in SSc skin, whereas *POSTN* was upregulated in SSc skin (Supplementary Fig S5D). Additionally, *SCARA5* exhibited distinct spatial enrichment patterns that were mutually exclusive to regions enriched for *POSTN* and *COMP* across multiple samples (Fig 3D; Supplementary Fig S5E). Estimation of module scores using marker genes for each subset revealed that Fib2 and Fib9 were spatially colocalised (Fig 3E; Supplementary Fig S5F). Moreover, we observed increased activities of JUN, SMAD3, and the TGF- β pathway, which were closely associated with the spatial localisation of Fib2 and Fib9 (Fig 3F,G; Supplementary Fig S5G,H). These findings aligned with their known profibrotic roles, further confirming the spatial mapping of these subclusters. Furthermore, we found that *SCARA5* expression negatively correlated with the mRSS, whereas *POSTN* expression positively correlated with mRSS (Fig 3H). Additionally, *SCARA5* showed an inverse correlation with platelet counts, while *POSTN* negatively correlated with non-RP disease duration. These results collectively suggested divergent effects of *SCARA5*⁺ and *POSTN*⁺ fibroblasts in SSc.

Differentiation of *SCARA5*⁺ to *POSTN*⁺ fibroblasts in SSc skin

Several studies have identified *SCARA5*⁺ fibroblasts as myofibroblast progenitors in fibrosis [28–30]. To determine whether this is true for SSc skin fibrosis, we integrated scRNA-seq data from mouse skin challenged by bleomycin (BLM) at various time points (days 0, 3, 7, 14, and 28) [35], identifying 14 major cell types and 9 fibroblast subclusters (Supplementary Fig S6A,B). Interestingly, we found a progressive decrease in *Scara5*⁺ fibroblasts following BLM treatment, while *Postn*⁺ fibroblasts substantially increased on days 14 and 28 (Supplementary Fig S6C,D). These temporal changes suggested a potential differentiation process from *Scara5*⁺ to *Postn*⁺ fibroblasts. Treatment of TGF- β or IL-1 β induced the downregulation of *SCARA5* and upregulation of *POSTN* and *ACTA2* in primary cultured human dermal fibroblasts (Supplementary Fig S6E).

Based on these observations, we next inferred 2 pseudotime trajectories from Fib1 to Fib2 (trajectory 1) and Fib1 to Fib9 (trajectory 2) (Fig 3I,J). We next estimated the expression variation of each gene along the pseudotime and identified 1524 and 1515 genes associated with trajectories 1 and 2, respectively. For trajectory 1, we found many genes were downregulated during the differentiation, including precursor marker *CD34* [36] and *SFRP2* [11] (Fig 3K; Supplementary Fig S6F). Moreover, we

identified a set of genes, including *C3*, *C7*, *CCL2*, and *CCL19*, which were particularly enriched at the intermediate stage of the trajectory, suggesting the involvement of chemokine and complement signalling during the differentiation process. Genes associated with the late stage of this trajectory included *FOS* and *JUNB*. For trajectory 2, we observed that *TIMP2*, *TGFBR3*, and *APOD* were additionally associated with the intermediate stage (Fig 3K; Supplementary Fig S6G). These results revealed shared and distinct gene regulatory networks that are involved in the divergent differentiation of fibroblasts.

Interestingly, Sinha et al [7] conducted comparative single-cell multiomics in reindeer following full-thickness injury and found that the velvet skin regenerates, while the back skin heals by forming a scar [7]. Similarly, in reindeer skin, *SCARA5* gradually decreased but *ACTA2* increased after wound in their back, but not in velvet (Supplementary Fig S7). This pattern is reminiscent of fibrotic tissue remodelling in human, which indicates the fibroblast dynamic changes could be conserved across species and tissues.

Spatial colocalisation of fibroblast subclusters and other cell types

To investigate spatial dependencies between fibroblast subsets and other major cell types, we conducted spatial colocalisation analyses in 2 distinct proximity areas: (1) co-occurrence within 1 single spot, determined by Spearman correlation between the module score of each fibroblast subcluster and the proportion of each major cell type, and (2) extended neighbourhood within an annular region encompassing up to 3 surrounding spots for each target spot analysed with the MISTy pipeline [37]. We focused specifically on Fib1, Fib2, and Fib9, as they have been demonstrated to exhibit significant altered in SSc (Fig 3C). We observed a consistent spatial dependency in both models across tissues, characterised by an increased co-occurrence between myofibroblast progenitor *SCARA5*⁺ Fib1 and macrophages, whereas *POSTN*⁺/*ACTA2*⁺ myofibroblasts (Fib2 and Fib9) were found in proximity to vascular smooth muscle cells (Fig 3L,M). These findings highlighted the potential critical role of macrophage in establishing the spatial microenvironment surrounding fibroblast from the early stage.

Correlations of the cell type niche and cellular phenotypes with clinical outcomes of SSc

We next associated the cell type niches with the clinical outcomes and treatment responsiveness in patients with SSc. All patients with SSc included in our study were under the standard treatment with mycophenolate mofetil (MMF) for consecutive 52 weeks. The proportion of niche 0, mainly comprising fibroblasts and macrophages, was significantly higher in the patients with SSc who were nonresponsive to a 52-week standard treatment of MMF, compared with those who were responsive (median proportion in responder, 18.28%, *n* = 5; median proportion in nonresponder, 25.00%, *n* = 5; *P* = .0079) (Fig 4A). The responsiveness was defined according to the criteria of the revised ACR-CRISS (American College of Rheumatology Composite Response Index in Systemic Sclerosis), which is the

vascular smooth muscle cells (VSMCs) and the state scores of Fib1 and Fib2 in a representative SSc skin sample. ALT, alanine aminotransferase; AST, aspartate aminotransferase; C4, complement component 4; CK, creatine kinase; Cr, creatinine; CRP, C-reactive protein; C3, complement component 3; DU, digital ulcer; ESR, erythrocyte sedimentation rate; EUSTAR, European Scleroderma Trials and Research Group; HGB, haemoglobin; IgG, immunoglobulin G; ILD, interstitial lung disease; LDH, lactate dehydrogenase; mRSS, modified Rodnan skin score; PLT, platelet count; γ -GT, gamma-glutamyl transferase; RP, Raynaud phenomenon; sPAP, systolic pulmonary arterial pressure; SpO₂, oxygen saturation; TB, total bilirubin; TnT, troponin T; WBC, white blood cell.

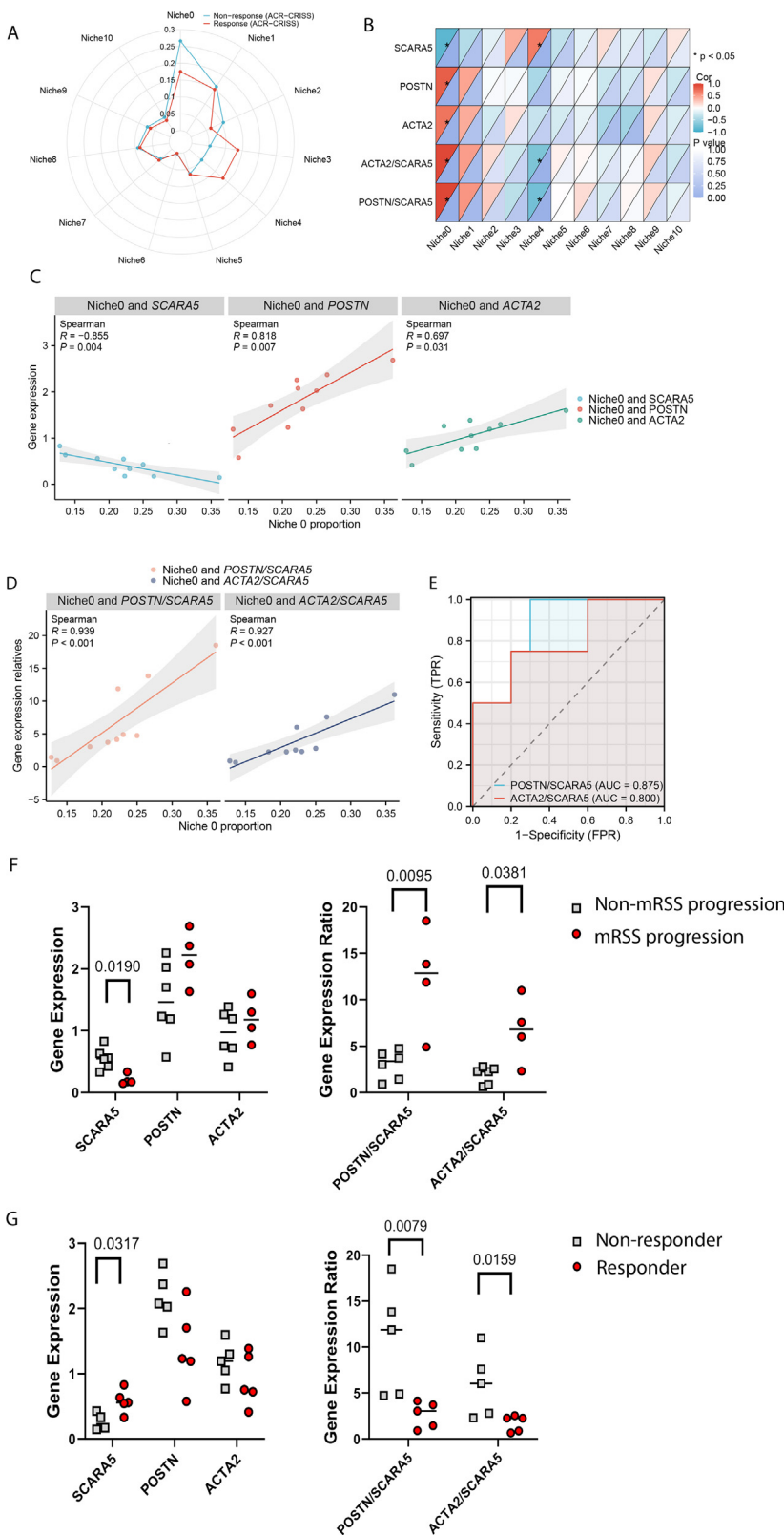


Figure 4. Correlations of the cellular phenotype with clinical outcomes. (A) A radar plot displaying the proportions of the identified niches in patients with systemic sclerosis (SSc), distinguishing between those who were responsive to the standard treatment of mycophenolate mofetil (MMF) and those who were not. (B) Heatmap showing correlations between the proportion of each niche and the pseudobulk gene expression, along with the ratios of *POSTN/SCARA5* and *ACTA2/SCARA5*. The colours in the upper left square represent Spearman r value, while those in the lower right square represent P values ($*P < .05$). (C, D) Correlations between the proportion of niche 0, which was predominantly composed of fibroblasts and macrophages, and the gene expressions of *SCARA5*, *POSTN*, and *ACTA2* (C), along with their ratios (D). Correlation data were analysed using nonparametric Spearman tests. Each data point represents 1 SSc sample, and the CI is indicated by grey shadows. (E) Receiver operating characteristic (ROC) curve showing the discrimination of patients with SSc from healthy controls based on the gene expression ratios of *POSTN/SCARA5* and *ACTA2/SCARA5*, respectively. The area under the curve (AUC) values are indicated in graphs. (F) The gene expressions of *SCARA5*, *POSTN*, and *ACTA2*, along with their ratios, comparing the patients with SSc who experienced skin fibrosis progression with those who were not. (G) The gene expression of *SCARA5*, *POSTN*, and *ACTA2*, along with their ratios, comparing the patients with SSc who were responsive to the standard treatment of MMF with those who were not, evaluated at a 52-week follow-up. Two-sided Wilcoxon rank-sum tests were conducted to compare the differences between the 2 groups.

improvement in ≥ 3 of 5 core clinical measurements with minimal clinically important difference of 25% for mRSS, 5% for FVC% predicted, 20% for health assessment questionnaire-disability index, and 20% for patient and clinician global assessments [38].

Consistently, the proportions of niche 0, enriched with fibroblasts and characterised by high activity of TGF- β signalling, correlated negatively with *SCARA5* and positively with *POSTN* and *ACTA2*, and more pronounced with the gene expression

ratios of *POSTN/SCARA5* and *ACTA2/SCARA5* (Fig 4B-D). Furthermore, the ratios of *POSTN/SCARA5* and *ACTA2/SCARA5* manifested the ability to discriminate the patients with SSc from healthy controls with the area under the curve of 0.875 and 0.800 ($P = .0339$ and $.0897$, respectively) (Fig 4E). Those baseline ratios were observed to be significantly increased in the patients who would experience skin fibrosis progression (Fig 4F), determined as over 25% increment in mRSS assessed at 52-week follow-up. Notably, the higher baseline ratios of

POSTN/SCARA5 and *ACTA2/SCARA5* were closely linked to the nonresponsiveness for the standard treatment with MMF for 52 weeks (Fig 4G). These data highlighted the potentials of aforementioned gene expression ratio in identification of the standard treatment–refractory patients with SSc, who may require more advanced treatment regimens rather than standard MMF.

To further validate our findings, we analysed the basal gene expression using bulk RNA-seq data from 6 patients with SSc who received various immunosuppressant and antifibrotic treatment (Supplementary Table S3). We observed an upwards trend in *POSTN/SCARA5* in patients with SSc with skin fibrosis progression and nonresponsiveness to treatment (Supplementary Fig S5I). After normalising and pooling RNA-seq data from 16 patients with SSc with different treatment backgrounds, we confirmed these trends, with higher *POSTN/SCARA5* levels in patients showing mRSS progression or treatment refractoriness (Supplementary Fig S5J). Similar results were seen in the genetics of inflammatory skin disease and systemic outcomes study (GENISOS) cohort (Supplementary Fig S5K; Supplementary Table S4). These data may provide insights into the future precision stratification strategy for patients with SSc.

Characterisation of macrophage heterogeneity

We next clustered the scRNA-seq data of macrophages and identified 3 distinct subsets (Fig 5A; Supplementary Fig S8A). We observed 1 cluster highly expressing proinflammatory cytokine *IL-1 β* , as well as *CXCL8*, *CCL3*, *CXCL2*, and *PTGS2*, thus termed as proinflammatory macrophages (Fig 5B). This subset also highly expressed *NR4A1*, *NR4A2*, and *NR4A3*, which have been identified as immediate early genes in response to various stimuli and have been linked to the pathogenesis of fibrotic tissue remodelling [39,40]. We also identified a subcluster characterised by phagocytic markers, including *MARCO*, *LYVE1*, *CD36*, *FCGRT*, and *C1QA*, therefore defined as phagocytic macrophages. The scavenging receptor CD36, known to mediate uptake in macrophage, has been implicated in the pathogenesis of tissue fibrosis [41,42]. *MARCO* was found to be upregulated in tissue-resident macrophages and circulating monocytes upon activation, and *MARCO*⁺ myeloid cells have been recognised as therapeutic targets for organ fibrosis in SSc [43,44]. *LYVE1*-expressing macrophage has been demonstrated to form coordinated multicellular nest structures that are distributed proximal to vasculature [45]. Another subset of macrophage that may be involved in the antigen presentation process is characterised by their overexpression of *HLA-B* and *HLA-C*, hereafter referred to as antigen-presenting macrophages. Indeed, the antigen-presenting macrophage and phagocytic macrophage defined in this study largely resembled previously identified *LYVE1*^{lo}*MHCII*^{hi} and *LYVE1*^{hi}*MHCII*^{lo} macrophages, which are conserved across tissues [10]. Notably, the *LYVE1*^{hi}*MHCII*^{lo} macrophages have been demonstrated to play a critical role in maintaining homeostasis, as depletion of *LYVE1*^{hi}*MHCII*^{lo} macrophages exacerbated experimental lung and heart fibrosis, along with exacerbated vessel permeability, immune infiltration, and collagen deposition [10].

The comparison of cell proportion between healthy and SSc samples showed that proinflammatory macrophages were significantly enriched in SSc, whereas the abundance of phagocytic macrophages was significantly decreased in SSc samples (Fig 5C). We also estimated TF activity for each cluster and observed a high activity of REL, RELA, and RELB in proinflammatory macrophages (Supplementary Fig S8B). These TFs are key members of the NF- κ B pathway that regulates immune and

inflammatory responses [46], further supporting the functions of proinflammatory macrophages.

Crosstalk between fibroblast and macrophage subclusters

Our analysis of spatial transcriptomic data revealed a strong spatial association between macrophages and fibroblasts, underscoring their potential functional interplay within the tissue microenvironment (Figs 2C and 3L). To further investigate the cellular crosstalk between specific macrophage and fibroblast subtypes, we performed a detailed cell–cell communication analysis using scRNA-seq data based on ligand–receptor (L–R) networks, identifying 393 significant interactions (Fig 5D). Among macrophage subsets, proinflammatory macrophages exhibited the highest number of interactions with fibroblast subsets ($n = 179$ interactions), suggesting that they are central players in fibroblast-mediated signalling (Fig 5E). Particularly, Fib9 emerged as a key fibroblast subtype with the strongest connectivity to macrophages ($n = 83$ interactions), outpacing other fibroblast subsets in its interaction capacity.

Visualisation of the specific L–R interactions revealed that macrophages receive signals from fibroblasts through multiple signalling pathways, including THBS, MIF, CXCL, LAMININ, and COLLAGEN (Supplementary Fig S8C). Among them, receptor CXCR4 was observed to interact with fibroblast-derived ligands such as MIF and CXCL12 (Fig 5F). This suggested a mechanism by which fibroblasts recruit and retain proinflammatory macrophages, thereby sustaining local inflammation and contributing to the progression of fibrosis. On the contrary, fibroblasts also acted as signal receivers, engaging the receptor ACKR3 to interact with MIF and CXCL12 (Fig 5G; Supplementary Fig S8D). Notably, *MIF* and *CXCL12* were expressed by nearly all subtypes of fibroblasts and macrophages (Fig 5H,I), underscoring their widespread involvement in cellular crosstalk.

However, the expression of CXCR4 was specific to proinflammatory macrophages, while *ACKR3* was uniquely expressed in the Fib1 myofibroblast progenitor subcluster (Fig 5H). A significant decrease in *ACKR3* expression in SSc fibroblasts has been observed (Supplementary Fig S8E,F). To confirm at the protein levels, we applied tyramide signal amplification (TSA) high-plex immunofluorescence onto adjacent tissue sections. Our TSA staining confirmed the overlapping expression of ACKR3 in SCARA5⁺ fibroblast (Fig 5J). Again, we observed *IL-1 β* ⁺ macrophages, which were also positive for CXCR4, closely distributed adjacent to fibroblasts (Fig 5K). These results revealed a highly organised and reciprocal signalling network between fibroblasts and macrophages, underscored by distinct L–R interactions. Of particular interest, the chemokine CXCL12 can bind to its classic receptor CXCR4, a G protein–coupled receptor, facilitating the recruitment of CXCR4⁺ cells, such as proinflammatory macrophages in our case. Alternatively, CXCL12 can interact with its atypical chemokine receptor ACKR3 independent of G protein–coupled receptor mediated signalling. ACKR3 is capable of actively internalising its chemokine ligands CXCL12, through scavenging. This function helps to modulate the availability and gradient of CXCL12 in the extracellular environment [47]. The selective expression of *ACKR3* in myofibroblast progenitors may indicate an endogenous compensation at an early stage. A dysfunction in this mechanism could lead to the subsequent CXCR4 mediated recruitment of macrophages, during the transition from myofibroblast progenitor to mature myofibroblast over the course of SSc progression. Pharmacologic inhibition of CXCR4 attenuated tissue fibrosis in bleomycin-induced

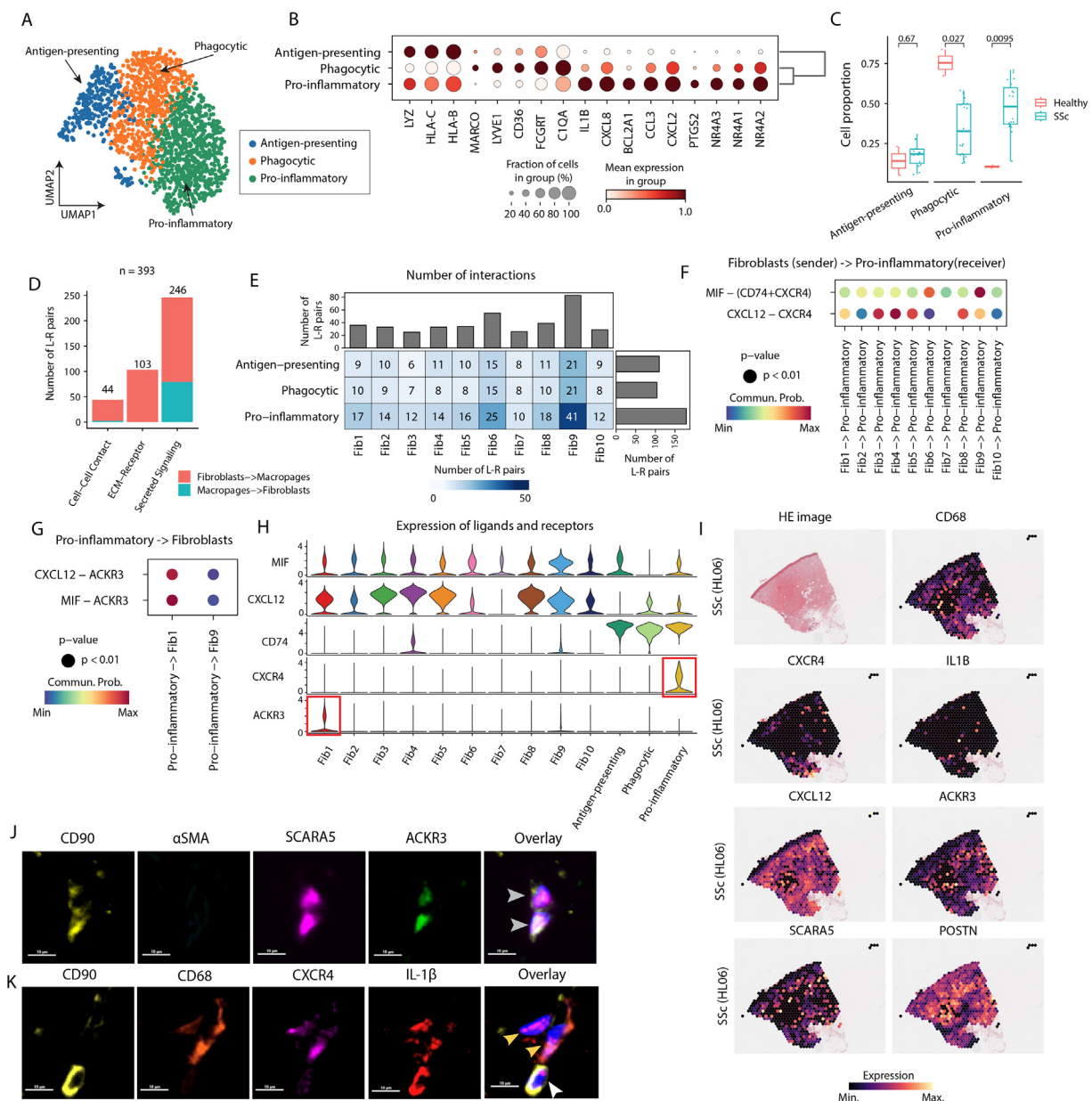


Figure 5. Characterisation of macrophage subclusters. (A) Uniform manifold approximation and projection (UMAP) showing subclusters of macrophages based on single-cell RNA sequencing (scRNA-seq) data. (B) Dotplot showing the marker genes for each subcluster. (C) Boxplot comparing the proportion of each subcluster between healthy and systemic sclerosis (SSc) samples. Each dot represents 1 sample. *P* values are calculated using the Wilcoxon rank-sum test (unpaired; 2 sides). (D,E) Number of significant ligand–receptor (L–R) pairs between the fibroblasts and macrophages. (F) Heatmap showing significant L–R pairs between fibroblast subsets and proinflammatory macrophages, with fibroblast serving as signalling senders and macrophage as receivers. The colours refer to the communication probability, and the sizes represent *P* values. (G) Heatmap showing significant L–R pairs between proinflammatory macrophages and fibroblast subsets (Fib1 and Fib9), as macrophage serving as signalling senders and fibroblast as receivers. (H) Violin plot showing the expression of ligands and receptors in scRNA-seq. (I) A representative hematoxylin and eosin (HE) staining and the expression of ligands and receptors in spatial space. (J) Representative images of tyramide signal amplification (TSA) staining on the expression of ACKR3 in SCAR5⁺ fibroblast, which are labelled with CD90 (indicated by grey arrows). (K) Representative images of TSA staining on the co-occurrence of fibroblasts (marked by CD90 and indicated by a white arrow) with interleukin (IL)-1 β ⁺ macrophages, which are also positive for CXCR4 (indicated by yellow arrows). The staining in (J) and (K) were performed on adjacent tissue sections to those used for $\times 10$ Visium. Horizontal scale bars represent 10 μ m.

experimental models, as evidenced by decreased dermal thickness in the skin, reduced Ashcroft scores in the lungs, and lower myofibroblast counts in both tissues (Fig 6A–D; Supplementary Fig S9A,B).

A data-driven model of SSc

Collectively, we proposed a data-driven and experimentally validated disease model of SSc that elucidated the mechanisms under investigation in this study (Fig 7). An imbalance in fibroblast subsets in SSc was characterised by a decrease in

SCAR5⁺ myofibroblast progenitors and an increase in POSTN⁺ and ACTA2⁺ terminally differentiated myofibroblasts. The decreased phagocytic macrophages (LYVE1^{hi}MHCII^{lo}) may contribute to the impairment of vessel integrity. The infiltration of IL-1 β ⁺ proinflammatory macrophages was observed actively interact with fibroblast subsets. These proinflammatory macrophages were overexpressed with CXCR4, which may mediate their recruitment by interacting with CXCL12. Additionally, the expression of ACKR3 on fibroblasts, particularly in progenitors under homeostatic conditions, may regulate the local levels CXCL12 by internalising the ligand,

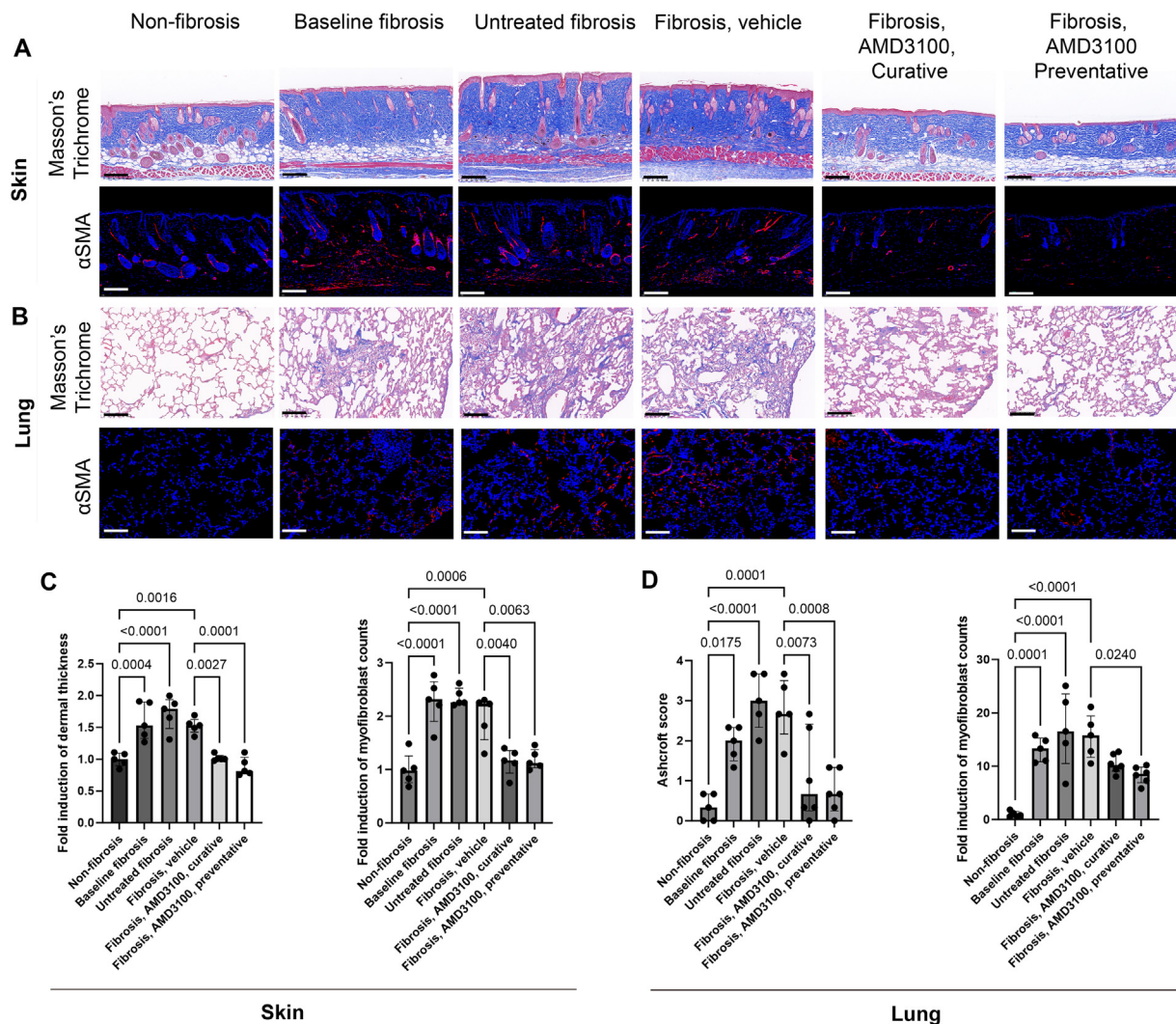


Figure 6. Pharmacologic inhibition of CXCR4 ameliorates experimental fibrosis models. The effect of CXCR4 inhibitor AMD3100 on experimental skin fibrosis (A,C) and lung fibrosis (B,D) induced by bleomycin is shown. (A) Representative Masson trichrome staining and immunofluorescent staining for the myofibroblast marker α SMA in skin sections are shown at $\times 100$ magnification (scale bars = 200 μ m). (B) Representative Masson trichrome staining and immunofluorescent staining for α -smooth muscle actin (SMA) in lung sections are shown at $\times 200$ magnification (scale bars = 100 μ m). (C) Quantification of dermal thickness and α SMA-positive myofibroblast in skin, normalised to control values. (D) Assessment of Ashcroft scores and quantification of α SMA-positive myofibroblast in the lungs, normalised to control values. Data in (C) and (D) are presented as median $\pm$ IQR from $n = 5$ -6 independent biological samples per group. P values determined by One-way analysis of variance (ANOVA) with Tukey multiple comparisons test as post hoc analysis.

thereby limiting excessive leukocyte recruitment and preventing unwanted tissue inflammation. However, this regulatory mechanism might diminish in effectiveness as myofibroblast progenitor differentiate to mature myofibroblast, ultimately resulting in uncontrolled CXCL12/CXCR4-mediated inflammatory cell attraction. This regulatory network fosters a shift from tissue homeostasis to a fibrogenic state, driving the progression of SSc. Therefore, the integration of spatial omics with scRNA-seq provides insight into the immune-stromal crosstalk involved in the pathogenesis of SSc.

DISCUSSION

In this study, we provided a comprehensive spatially resolved map of the skin ecosystem in SSc by integrating spatial transcriptomics with scRNA-seq data. Our results revealed a disease-specific microenvironment enriched with fibroblasts and macrophages exhibiting high TGF- β signalling pathway activity and strong correlation with clinical outcomes in SSc. Moreover, we uncovered a proportional imbalance of fibroblast subsets in

SSc, with a significant reduction in $SCARA5^+$ fibroblasts and an upregulation in $POSTN^+/ACTA2^+$ fibroblasts compared with healthy controls. Of note, $SCARA5^+$ fibroblasts have been recently reported as myofibroblast progenitors in heart and kidney [28,29,48]. In addition, a cross-tissue human fibroblast atlas also demonstrated the substantial overlaps among $SCARA5^+$ fibroblasts and progenitor-like fibroblasts, as well as between $POSTN^+/ACTA2^+$ fibroblasts and myofibroblasts [49]. These findings suggest that the differentiation from $SCARA5^+$ myofibroblast progenitors to $POSTN^+/ACTA2^+$ myofibroblasts may represent a conserved fibrotic mechanism across different tissues, organs, and diseases. Targeting the shared mechanism has emerged as a strategy for basket trial design especially for rare diseases that share the same triggering or perpetuating signalling pathways [50].

Of note, we characterised the intricate interplay of fibroblasts and macrophages within niches that potentially drive the transition from a homeostatic to a fibrogenic state. Further investigation into cell–cell communications offered potential for identifying targetable disease-driving molecules for therapeutic

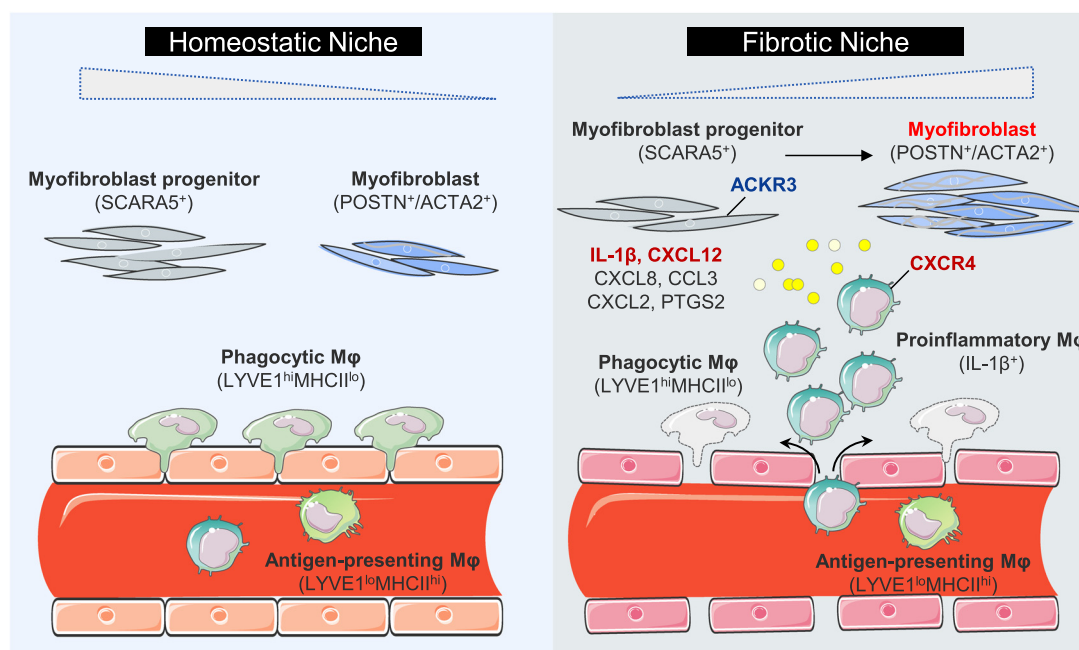


Figure 7. A data-driven model of systemic sclerosis (SSc). Schematic summary illustrates the proposed cellular interplay in SSc skin, which leads to a shift from tissue homeostasis to a fibrogenic environment. The illustration was created by using images from Servier Medical Art (<http://smart.servier.com/>), licensed under CC BY 3.0 (<https://creativecommons.org/licenses/by/3.0/>).

intervention. We revealed an enhanced interaction between IL-1 β ⁺ proinflammatory macrophages and fibroblast subsets in SSc. Infiltrating macrophages activate fibroblasts by secreting IL-1 β , while systemic IL-1 β inhibition or targeted IL-1 β deletion in macrophages prevents fibroblast activation [3,4,51]. The role of IL-1 family of cytokines, including IL-1, IL-18, IL-33, and IL-36, has been implicated in the pathogenesis of SSc [3,4,51,52]. For instance, the elevated IL-36, which has been reported to mediate cellular crosstalk in SSc, represents a promising area for further research [52].

Furthermore, we found this crosstalk was predicted to be mediated by the interaction of CXCL12 with CXCR4 in macrophages and ACKR3 in myofibroblast progenitors. Moreover, we also found the deregulation of LYVE1^{hi}MHCII^{lo} macrophages in SSc, which are known to play a protective role by maintaining vascular integrity and preventing immune cell infiltration in fibrotic conditions [10]. These sequential events such as tissue entry and immune infiltration occurring in SSc may shift the hemostatic condition towards a profibrotic environment. Importantly, the inability to constrain CXCL12 overexpression in the milieu through internalisation by myofibroblast progenitors via ACKR3 leads to CXCR4-mediated proinflammatory macrophage recruitment, which is reflective of the fine-tuned regulation by local fibroblasts.

Translating omics discoveries into clinically actionable strategies remains a critical unmet challenge in SSc. Our research contributes novel insights to clinical practice in biomarker-driven patient stratification and mechanism-based target identification:

Firstly, we found the gene ratio of *POSTN*/*SCARA5* emerges as a predictive indicator for the clinical progression of skin fibrosis and treatment refractoriness in SSc, which actually aligned and reinforced the predictive values of basal molecular pattern. The study by Hinchcliff et al [53] showed that 571 genes (false discovery rate < 10%) had changed between pre-MMF and post-MMF treatment biopsies for patients demonstrating mRSS improvement [53]. The study by Assassi et al [54] showed the 12-month treatment of MMF yield changes in peripheral blood gene expression profiling, highlighting the predictive

significance of baseline peripheral blood cell transcript scores for response to MMF treatment in SSc–interstitial lung disease [54]. We additionally validated and extended these data in the patients with SSc with different treatment background, highlighting the potential of using gene expression profiling in personalised stratification strategies.

Secondly, we identified that the interplay between fibroblasts and macrophages may drive the transition from homeostasis to a profibrotic environment. Targeting the key regulators within this network may abrogate the profibrotic transition and ameliorate disease. We validated our data-driven model by inhibiting CXCR4 in 2 preclinical murine fibrosis models. Our study demonstrated a promising approach to translating omics findings into therapeutic target identification.

Beyond the molecular signature, restoring the disorganised distribution in spatial space might represent a new dimension in pathogenesis exploration and therapy development [55]. Likewise, we found a loss of the polarised distribution of fibroblast subsets in SSc skin shown in Figure 3D. In healthy skin, the *SCARA5*⁺ Fib1 are predominantly located in the papillary dermis in comparison with the reticular dermis, while *POSTN*⁺ Fib2 are scarcely observed in the dermis and mainly in the upper layer. In SSc, both *SCARA5* and *POSTN* shift to a reticular expression pattern. Therefore, this phenomenon indicates the need in the investigation into the mobility and migration of fibroblasts during tissue remodelling as the future effort. Genome-wide spatial omics exhibited the advantage in characterising this physiologic skin organisation.

While spatial transcriptomics provides a unique advantage in analysing the structured tissue niches, our study was limited by the relatively low resolution of the 10× Visium platform (55- μ m per spot). To mitigate this limitation, we leveraged Stereo-seq, an emerging spatial transcriptomics technology with an unprecedented resolution of 0.5 μ m per spot. To achieve a comprehensive description of the skin landscape in SSc, we also recognised that 3-dimensional profiling of tissue space and subcellular localisation of protein signals will significantly expand our understanding in SSc pathogenesis.

We also acknowledge the limitations of bleomycin-induced fibrosis models applied in our study. We selected this model since we primarily focused on the immune-stromal crosstalk in SSc, as Sargent et al [56] showed that cutaneous gene expression patterns in bleomycin-induced skin fibrosis overlapped with the inflammatory intrinsic gene subset of human patients with SSc [56]. While they mimic early inflammatory and fibrotic phases of SSc, they are limited to the injection site and primarily driven by inflammation, which is not entirely consistent with human SSc. Subcutaneous injection of bleomycin will not affect lung remarkably; therefore, we also included a bleomycin-induced lung fibrosis mouse model to compensate for this limitation. Moreover, the potential immunostimulatory effects of AMD3100 through bone marrow stem cell mobilisation, which indeed complicate translation from murine models to human disease. However, this effect is transient, reversible, and tolerable in human [57]. Inhibition of CXCR4 using AMD3100 exhibited antifibrotic effect in multiple preclinical fibrosis models [58–60]. Further optimising the route and dosage of AMD3100 administration—such as through localised delivery—could minimise unintended stem cell mobilisation while maximising therapeutic efficacy.

In summary, this study reveals a fibroblast–macrophage connectivity network underlying SSc pathogenesis by integrating spatial multiomics with scRNA-seq data. Within this network, fibroblasts and macrophages with distinct phenotypes contribute to tissue fibrotic remodelling in a coordinated manner. This approach enables the identification of key regulatory interactions and potential therapeutic targets for fibrosis intervention.

Competing interests

Although no disclosures directly relate to the present study, JHWD has consultancy relationships with Active Biotech, Anamar, ARXX, AstraZeneca, Bayer Pharma, Boehringer Ingelheim, Callidatas, Celgene, Galapagos, GSK, Inventiva, Janssen, Kyverna, Novartis, Pfizer, Quell Therapeutics, and UCB; has received research funding from Anamar, ARXX, BMS, Bayer Pharma, Boehringer Ingelheim, Cantargia, Celgene, CSL Behring, Exo Therapeutics, Galapagos, GSK, Incyte, Inventiva, Kiniksa, Kyverna, Lassen Therapeutics, Mestag, Sanofi-Aventis, RedX, UCB, and ZenasBio; and is the CEO of 4D Science and scientific lead of FibroCure. LP has financial interests in Edilytics Inc, and SeQure Dx Inc. All other authors declare they have no competing interests.

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Contributors

ML and ZL were responsible for conceptualisation of the study. ZL, ML, ARR, WX, LH, WY, Y-NL, A-EM, CB and XS were responsible for methodology. ZL, ML, and ARR performed investigations and visualised the study. ML, WY, LH, and JW were responsible for resources. ML, JHWD, LP, and HZ were responsible for funding acquisition. ML administered the project. ML, ZL, JHWD, and RH supervised the study. ML and ZL wrote the

original draft. All authors reviewed, edited, read, and approved the final manuscript.

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Patient consent for publication

All patients and controls signed an informed consent form approved by the local institutional review board.

Ethics approval

All human studies were approved by the ethical committees of Huashan Hospital of Fudan University.

Provenance and peer review

Not commissioned; externally peer reviewed.

Data availability statement

The spatial transcriptomics data presented in this study are available at the Zenodo data archive (<https://zenodo.org/records/14577696>). The scRNA-seq data for human skin were downloaded from GEO with the accession number GSE195452 (Gur2022) and GSE138669 (Tabib2021). A cross-tissue human fibroblast atlas was analysed with the interface website <http://pan-fib.cancer-pku.cn> [49]. The scRNA-seq data of full-thickness injuries in skin from reindeer was analysed with the interface website w.biernaskielab.ca/reindeer_atlas [7]. The scRNA-seq data for mouse skin were downloaded from the Zenodo data archive (<https://zenodo.org/records/7455989>). The codes used in this study are available on Github: https://github.com/lzj1769/Spatial_SSc_project

Supplementary materials

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.jard.2025.04.025](https://doi.org/10.1016/j.jard.2025.04.025).

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REFERENCES

- [1] Elhai M, Meune C, Boubaya M, Avouac J, Hachulla E, Balbir-Gurman A, et al. Mapping and predicting mortality from systemic sclerosis. *Ann Rheum Dis* 2017;76:1897–905.
- [2] Distler JHW, Györfi A-H, Ramanujam M, Whitfield ML, Königshoff M, Lafyatis R. Shared and distinct mechanisms of fibrosis. *Nat Rev Rheumatol* 2019;15:705–30.
- [3] Alexanian M, Padmanabhan A, Nishino T, Travers JG, Ye L, Pelonero A, et al. Chromatin remodelling drives immune cell-fibroblast communication in heart failure. *Nature* 2024;635:434–43.
- [4] Amrute JM, Luo X, Penna V, Yang S, Yamawaki T, Hayat S, et al. Targeting immune-fibroblast cell communication in heart failure. *Nature* 2024;635:423–33.
- [5] Mayr CH, Santacruz D, Jarosch S, Bleck M, Dalton J, McNabola A, et al. Spatial transcriptomic characterization of pathologic niches in IPF. *Sci Adv* 2024;10:eadi5473.
- [6] Morse C, Tabib T, Sembrat J, Buschur KL, Bittar HT, Valenzi E, et al. Proliferating SPP1/MERTK-expressing macrophages in idiopathic pulmonary fibrosis. *Eur Respir J* 2019;54:1802441.
- [7] Sinha S, Sparks HD, Labit E, Robbins HN, Gowing K, Jaffer A, et al. Fibroblast inflammatory priming determines regenerative versus fibrotic skin repair in reindeer. *Cell* 2022;185 4717–36.e25.
- [8] Brown M, O, Reilly S. The immunopathogenesis of fibrosis in systemic sclerosis. *Clin Exp Immunol* 2019;195:310–21.
- [9] Worrell JC, O'Reilly S. Bi-directional communication: conversations between fibroblasts and immune cells in systemic sclerosis. *J Autoimmun* 2020;113:102526.
- [10] Chakarov S, Lim HY, Tan L, Lim SY, See P, Lum J, et al. Two distinct interstitial macrophage populations coexist across tissues in specific subcutaneous niches. *Science* 2019;363 eaau0964.
- [11] Tabib T, Huang M, Morse N, Papazoglou A, Behera R, Jia M, et al. Myofibroblast transcriptome indicates SFRP2hi fibroblast progenitors in systemic sclerosis skin. *Nat Commun* 2021;12:4384.
- [12] Gur C, Wang S-Y, Sheban F, Zada M, Li B, Kharouf F, et al. LGR5 expressing skin fibroblasts define a major cellular hub perturbed in scleroderma. *Cell* 2022;185 1373–88.e20.
- [13] Clark KEN, Xu S, Attah M, Ong VH, Buckley CD, Denton CP. Single-cell analysis reveals key differences between early-stage and late-stage systemic sclerosis skin across autoantibody subgroups. *Ann Rheum Dis* 2023;82:1568–79.
- [14] Yin H, Distler O, Shen L, Xu X, Yuan Y, Li R, et al. Endothelial response to type I interferon contributes to vasculopathy and fibrosis and predicts disease progression of systemic sclerosis. *Arthritis Rheumatol* 2024;76:78–91.
- [15] Ma F, Tsou P-S, Gharaee-Kermani M, Plazyo O, Xing X, Kirma J, et al. Systems-based identification of the Hippo pathway for promoting fibrotic mesenchymal differentiation in systemic sclerosis. *Nat Commun* 2024;15:210.
- [16] Rius Rigau A, Liang M, Devakumar V, Neelagar R, Matei AE, Györfi AH, et al. Imaging mass cytometry-based characterisation of fibroblast subsets and their cellular niches in systemic sclerosis. *Ann Rheum Dis* 2024;ard-2024-226336. doi: 10.1136/ard-2024-226336.
- [17] Rius Rigau A, Li Y-N, Matei A-E, Györfi AH, Bruch PM, Koziel S, et al. Characterization of vascular niche in systemic sclerosis by spatial proteomics. *Circ Res* 2024;134:875–91.
- [18] Chen A, Liao S, Cheng M, Ma K, Wu L, Lai Y, et al. Spatiotemporal transcriptomic atlas of mouse organogenesis using DNA nanoball-patterned arrays. *Cell* 2022;185:1777–1792.e21.
- [19] Dees C, Pötter S, Zhang Y, Bergmann C, Zhou X, Lubert M, et al. TGF- β -induced epigenetic deregulation of SOCS3 facilitates STAT3 signaling to promote fibrosis. *J Clin Invest* 2020;130:2347–63.
- [20] Bryon J, Wasson CW, Koeppen K, Chandler F, Willis LF, Di Donato S, et al. Systemic sclerosis dermal fibroblast exosomes trigger type 1 interferon responses in keratinocytes via a TBK/JAK/STAT signaling axis. *Arthritis Rheumatol* 2025;77:322–34.
- [21] Chakraborty D, Šumová B, Mallano T, Chen CW, Distler A, Bergmann C, et al. Activation of STAT3 integrates common profibrotic pathways to promote fibroblast activation and tissue fibrosis. *Nat Commun* 2017;8:1130.
- [22] Wang W, Bhattacharyya S, Marangoni RG, Carns M, Dennis-Aren K, Yeldandi A, et al. The JAK/STAT pathway is activated in systemic sclerosis and is effectively targeted by tofacitinib. *J Scleroderma Relat Disord* 2020;5:40–50.
- [23] Khanna D, Denton CP, Jhreis A, van Laar JM, Frech TM, Anderson ME, et al. Safety and efficacy of subcutaneous tocilizumab in adults with systemic sclerosis (faSScinate): a phase 2, randomised, controlled trial. *Lancet* 2016;387:2630–40.
- [24] Khanna D, Lin CJF, Furst DE, Goldin J, Kim G, Kuwana M, et al. Tocilizumab in systemic sclerosis: a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Respir Med* 2020;8:963–74.
- [25] Denton CP, Ong VH, Xu S, Chen-Harris H, Modrusan Z, Lafyatis R, et al. Therapeutic interleukin-6 blockade reverses transforming growth factor- β pathway activation in dermal fibroblasts: insights from the faSScinate clinical trial in systemic sclerosis. *Ann Rheum Dis* 2018;77:1362–71.
- [26] Avouac J, Palumbo K, Tomcik M, Zerr P, Dees C, Horn A, et al. Inhibition of activator protein 1 signaling abrogates transforming growth factor β -mediated activation of fibroblasts and prevents experimental fibrosis. *Arthritis Rheum* 2012;64:1642–52.
- [27] Bergmann C, Chenguiti Fakhouri S, Trinh-Minh T, Filla T, Rius Rigau A, Ekici AB, et al. Mutual amplification of GLI2/Hedgehog and transcription factor JUN/AP-1 signaling in fibroblasts in systemic sclerosis: potential implications for combined therapies. *Arthritis Rheumatol* 2025;77:92–8. doi: 10.1002/art.42979.
- [28] Kuppe C, Ibrahim MM, Kranz J, Zhang X, Ziegler S, Perales-Patón J, et al. Decoding myofibroblast origins in human kidney fibrosis. *Nature* 2021;589:281–6.
- [29] Kuppe C, Ramírez Flores RO, Li Z, Hayat S, Levinson RT, Liao X, et al. Spatial multi-omic map of human myocardial infarction. *Nature* 2022;608:766–77.
- [30] Li Z, Kuppe C, Ziegler S, Cheng M, Kabgani N, Menzel S, et al. Chromatin-accessibility estimation from single-cell ATAC-seq data with scOpen. *Nat Commun* 2021;12:6386.
- [31] Nanri Y, Nunomura S, Terasaki Y, Yoshihara T, Hirano Y, Yokosaki Y, et al. Cross-talk between transforming growth factor- β and periostin can be targeted for pulmonary fibrosis. *Am J Respir Cell Mol Biol* 2020;62:204–16.
- [32] Nanri Y, Nunomura S, Honda Y, Takedomi H, Yamaguchi Y, Izuhara K. A positive loop formed by SOX11 and periostin upregulates TGF- β signals leading to skin fibrosis. *J Invest Dermatol* 2023;143 989–98.e7.
- [33] Liang R, Šumová B, Cordazzo C, Mallano T, Zhang Y, Wohlfahrt T, et al. The transcription factor GLI2 as a downstream mediator of transforming growth factor- β -induced fibroblast activation in SSc. *Ann Rheum Dis* 2017;76:756–64.
- [34] Skaug B, Khanna D, Swindell WR, Hinchcliff ME, Frech TM, Steen VD, et al. Global skin gene expression analysis of early diffuse cutaneous systemic sclerosis shows a prominent innate and adaptive inflammatory profile. *Ann Rheum Dis* 2020;79:379–86.
- [35] Liang Y, Hu Y, Zhang J, Song H, Zhang X, Chen Y, et al. Dynamic pathological analysis reveals a protective role against skin fibrosis for TREM2-dependent macrophages. *Theranostics* 2024;14:2232–45.
- [36] Nazari B, Rice LM, Stifano G, Barron AMS, Wang YM, Korndorf T, et al. Altered dermal fibroblasts in systemic sclerosis display podoplanin and CD90. *Am J Pathol* 2016;186:2650–64.
- [37] Tanevski J, Flores ROR, Gabor A, Schapiro D, Saez-Rodriguez J. Explainable multiview framework for dissecting spatial relationships from highly multiplexed data. *Genome Biol* 2022;23:97.
- [38] Khanna D, Huang S, Lin CJF, Spino C. New composite endpoint in early diffuse cutaneous systemic sclerosis: revisiting the provisional American College of Rheumatology composite response index in systemic sclerosis. *Ann Rheum Dis* 2021;80:641–50.
- [39] Boulet S, Le Corre L, Odagiu L, Labrecque N. Role of NR4A family members in myeloid cells and leukemia. *Curr Res Immunol* 2022;3:23–36.
- [40] Palumbo-Zerr K, Zerr P, Distler A, Fliehr J, Mancuso R, Huang J, et al. Orphan nuclear receptor NR4A1 regulates transforming growth factor- β signaling and fibrosis. *Nat Med* 2015;21:150–8.
- [41] Kwak D, Bradley PB, Subbotina N, Ling S, Teitz-Tennenbaum S, Osterholzer JJ, et al. CD36/Lyn kinase interactions within macrophages promotes pulmonary fibrosis in response to oxidized phospholipid. *Respir Res* 2023;24:314.
- [42] Pennathur S, Pasichnyk K, Bahrami NM, Zeng L, Febbraio M, Yamaguchi I, et al. The macrophage phagocytic receptor CD36 promotes fibrogenic pathways on removal of apoptotic cells during chronic kidney injury. *Am J Pathol* 2015;185:2232–45.
- [43] Papazoglou A, Huang M, Bulik M, Lafyatis A, Tabib T, Morse C, et al. Epigenetic regulation of profibrotic macrophages in systemic sclerosis-associated interstitial lung disease. *Arthritis Rheumatol* 2022;74:2003–14.
- [44] Xu D, Bhattacharyya S, Wang W, Ifergan I, Wong MAC, Procissi D, et al. PLG nanoparticles target fibroblasts and MARCO+ monocytes to reverse multiorgan fibrosis. *JCI Insight* 2022;7:e151037. doi: 10.1172/jci.insight.151037.
- [45] Anstee JE, Feehan KT, Opzoomer JW, Dean I, Muller HP, Bahri M, et al. LYVE-1+ macrophages form a collaborative CCR5-dependent perivascular niche that influences chemotherapy responses in murine breast cancer. *Dev Cell* 2023;58 1548–61.e10.
- [46] Liu T, Zhang L, Joo D, Sun SC. NF- κ B signaling in inflammation. *Signal Transduct Target Ther* 2017;2:17023. doi: 10.1038/sigtrans.2017.23.
- [47] Cuesta-Margolles G, Schlecht-Louf G, Bachelier F. ACKR3 in skin homeostasis, an overlooked player in the CXCR4/CXCL12 axis. *J Invest Dermatol* 2025;145:1039–49. doi: 10.1016/j.jid.2024.08.022.
- [48] Schreiber F, Anslinger TM, Kramann R. Fibrosis in pathology of heart and kidney: from deep RNA-sequencing to novel molecular targets. *Circ Res* 2023;132:1013–33.

- [49] Gao Y, Li J, Cheng W, Diao T, Liu H, Bo Y, et al. Cross-tissue human fibroblast atlas reveals myofibroblast subtypes with distinct roles in immune modulation. *Cancer Cell* 2024;42:1764–83.e10.
- [50] Flaherty KR, Wells AU, Cottin V, Devaraj A, Walsh SLF, Inoue Y, et al. Nintedanib in progressive fibrosing interstitial lung diseases. *N Engl J Med* 2019;381:1718–27.
- [51] Grönberg C, Rattik S, Tran-Manh C, Zhou X, Rius Rigau A, Li YN, et al. Combined inhibition of IL-1, IL-33 and IL-36 signalling by targeting IL1RAP ameliorates skin and lung fibrosis in preclinical models of systemic sclerosis. *Ann Rheum Dis* 2024;83:1156–68.
- [52] O'Reilly S. Interleukin-36 α is elevated in diffuse systemic sclerosis and may potentiate fibrosis. *Cytokine* 2022;156:155921.
- [53] Hinchcliff M, Huang C-C, Wood TA, Matthew Mahoney J, Martyanov V, Bhat-tacharyya S, et al. Molecular signatures in skin associated with clinical improvement during mycophenolate treatment in systemic sclerosis. *J Invest Dermatol* 2013;133:1979–89.
- [54] Assassi S, Volkmann ER, Zheng WJ, Wang X, Wilhalme H, Lyons MA, et al. Peripheral blood gene expression profiling shows predictive significance for response to mycophenolate in systemic sclerosis-related interstitial lung disease. *Ann Rheum Dis* 2022;81:854–60.
- [55] Fakhouri SC, Zhu H, Li YN, Ronicke M, Rigau AR, Dees C, et al. Disturbed spatial WNT activation-a potential driver of the reticularized skin phenotype in systemic sclerosis. *Arthritis Rheumatol* 2024 Published online December 25. doi: 10.1002/art.43094.
- [56] Sargent JL, Li Z, Aliprantis AO, Greenblatt M, Lemaire R, Wu MH, et al. Identification of optimal mouse models of systemic sclerosis by interspecies comparative genomics. *Arthritis Rheumatol* 2016;68:2003–15.
- [57] Hendrix CW, Flexner C, MacFarland RT, Giandomenico C, Fuchs EJ, Redpath E, et al. Pharmacokinetics and safety of AMD-3100, a novel antagonist of the CXCR-4 chemokine receptor, in human volunteers. *Antimicrob Agents Chemother* 2000;44:1667–73.
- [58] Song JS, Kang CM, Kang HH, Yoon HK, Kim YK, Kim KH, et al. Inhibitory effect of CXC chemokine receptor 4 antagonist AMD3100 on bleomycin induced murine pulmonary fibrosis. *Exp Mol Med* 2010;42:465–72.
- [59] Makino H, Aono Y, Azuma M, Kishi M, Yokota Y, Kinoshita K, et al. Antifibrotic effects of CXCR4 antagonist in bleomycin-induced pulmonary fibrosis in mice. *J Med Invest* 2013;60:127–37.
- [60] Ullah A, Chen G, Yibang Z, Hussain A, Shafiq M, Raza F, et al. A new approach based on CXCR4-targeted combination liposomes for the treatment of liver fibrosis. *Biomater Sci* 2022;10:2650–64.